

## AI Prompt Refining

### 1. Objective

In this practical, learners will understand how to improve a weak AI prompt into a clear, structured, and useful prompt.

By the end of this walkthrough, learners will be able to:

- Identify problems in vague prompts.
  - Use prompt refining techniques to improve AI responses.
  - Apply prompt refining to basic cybersecurity awareness tasks.
  - Compare weak and refined prompt outputs.
  - Use OpenRouter free models for basic cybersecurity learning activities.
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### 2. Lab Environment

For this practical, the following environment will be used:

- **Operating System:** Kali Linux
- **Browser:** Firefox
- **AI Platform:** OpenRouter
- **Models:** OpenRouter free models only
- **Paid models:** Not required
- **File uploads:** Not used in this practical
- **Use case:** Defensive cybersecurity awareness

Learners should already be logged in to OpenRouter before starting this practical.

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### 3. What is Prompt Refining?

**Prompt refining** means improving a basic or weak prompt so that the AI gives a better, clearer, and more useful answer.

A weak prompt is usually short and unclear.

A refined prompt gives the AI more direction by adding details such as:

- The role the AI should take
- The exact task to perform
- The target audience
- The context
- The constraints

- The desired output format

In simple words, prompt refining helps the AI understand **what you want, how you want it, and what it should avoid**.

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#### 4. Why Prompt Refining Matters in Cybersecurity

In cybersecurity, unclear prompts can produce incomplete, risky, or overly general answers.

For example, if a learner asks:

Explain phishing.

The AI may give a general answer, but it may not explain phishing in a way that is suitable for beginners or awareness training.

A better prompt can guide the AI to produce a safer and more useful response.

Prompt refining matters in cybersecurity because it helps:

- Get more accurate defensive explanations.
- Avoid unsafe or offensive details.
- Focus the output on awareness, detection, prevention, or response.
- Create structured outputs for reports, training, SOC analysis, or user education.
- Reduce confusion for beginners.

In cybersecurity training, a good prompt should clearly tell the AI to stay defensive and avoid harmful instructions.

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#### 5. A Weak Cybersecurity Prompt

For this practical, start with the following weak prompt:

Explain phishing.

This prompt is not wrong, but it is too broad.

It does not tell the AI:

- Who the explanation is for
- How detailed the answer should be
- Whether the answer should be offensive or defensive
- What points should be covered
- What format should be used

Because of this, the output may be generic or inconsistent.

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## 6. Testing the Weak Prompt on a Free OpenRouter Model

### Step 1: Open OpenRouter

Open Firefox in Kali Linux and go to the OpenRouter website.

Log in to your OpenRouter account if required.

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### Step 2: Open the Chat Interface

From the OpenRouter dashboard, navigate to the chat or model testing area.

Use a **free model** only.

We will use the GLM 4.5 Air (Free) for this exercise.

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### Step 3: Enter the Weak Prompt

In the prompt box, enter:

Explain phishing.

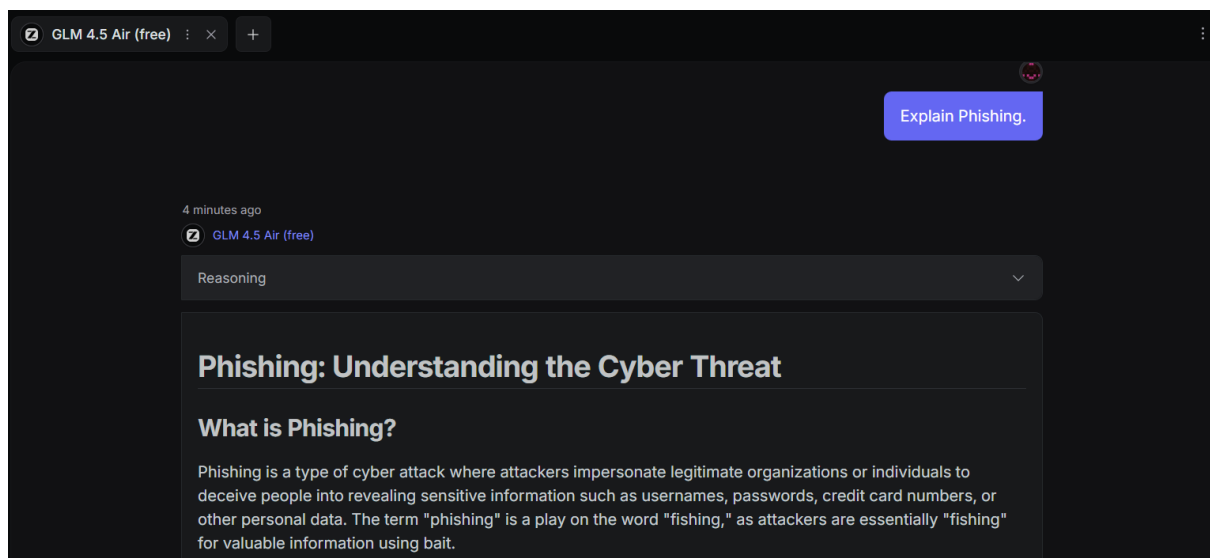
Submit the prompt.

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### Step 4: Read the Output

Carefully read the AI response.

The response will usually explain phishing, but it may be very general.



## 7. Observing Problems in the Output

After testing the weak prompt, learners should observe the response and identify possible issues.

Common problems may include:

- The explanation may be too short.
- The language may not be beginner-friendly.
- The answer may not include warning signs.
- The answer may not include safe actions.
- The answer may not be structured properly.
- The answer may not clearly focus on cybersecurity awareness.
- The AI may include unnecessary technical details.
- The output format may be inconsistent.
- **Most importantly, output is not aligned with what we may exactly require.**

The weak prompt gives the AI too much freedom.

That is why the result may not fully match the learning objective.

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## 8. Refining the Prompt

Now improve the prompt using five important prompt refining elements:

1. Role
  2. Task
  3. Context
  4. Constraints
  5. Output format
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### A. Role

The **role** tells the AI what identity or perspective it should use.

Example:

You are a cybersecurity awareness trainer.

This tells the AI to respond like someone teaching users about cybersecurity awareness.

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### B. Task

The **task** tells the AI exactly what to do.

Example:

Explain phishing to beginner-level users in simple language.

This is better than just saying:

Explain phishing.

The refined task gives a clear audience and style.

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### **C. Context**

The **context** tells the AI what situation the answer is meant for.

In this case, the context is beginner-level cybersecurity awareness.

Example:

Explain phishing to beginner-level users in simple language.

This helps the AI avoid overly technical explanations.

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### **D. Constraints**

**Constraints** tell the AI what it should not do.

This is very important in cybersecurity.

Example:

Constraints:

- Do not include offensive techniques.
- Do not provide instructions for creating phishing emails.
- Keep the explanation defensive and awareness-focused.

These constraints help keep the output safe and suitable for a cybersecurity awareness lab.

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### **E. Output Format**

The **output format** tells the AI how the answer should be organized.

Example:

Output format:

Use clear headings and bullet points.

This makes the answer easier to read and compare.

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## **9. Refined Cybersecurity Prompt**

Now use the refined prompt below:

You are a cybersecurity awareness trainer.

Explain phishing to beginner-level users in simple language.

Include:

1. What phishing means
2. Common warning signs of phishing emails
3. Why phishing is dangerous
4. Three safe actions users should take when they receive a suspicious email
5. One short example of a suspicious email subject line

Constraints:

- Do not include offensive techniques.
- Do not provide instructions for creating phishing emails.
- Keep the explanation defensive and awareness-focused.

Output format:

Use clear headings and bullet points.

This prompt is much better because it clearly defines:

- The role of the AI
  - The learning audience
  - The exact content required
  - The safety boundaries
  - The response format
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## **10. Testing the Refined Prompt**

### **Step 1: Stay in the Same OpenRouter Chat**

Use the same free model selected earlier.

If the earlier model has reached a rate limit, select another available free model.

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### **Step 2: Enter the Refined Prompt**

Copy and paste the refined prompt into the chat box.

Submit the prompt.

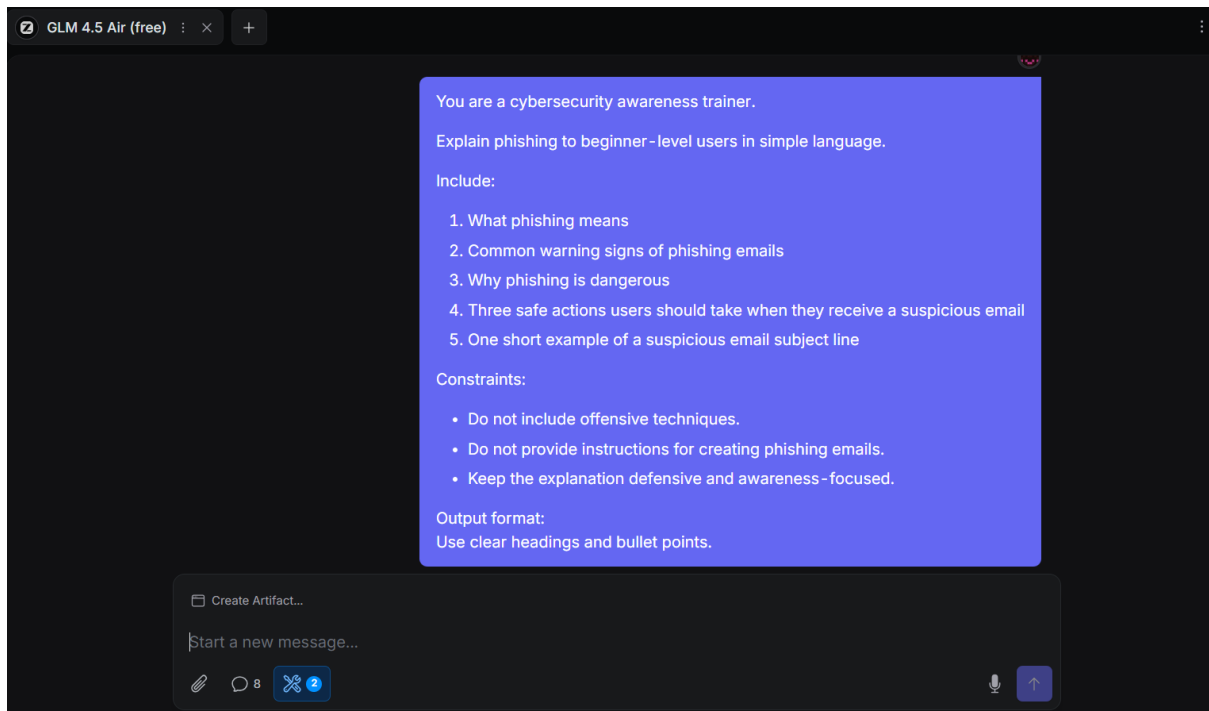
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### **Step 3: Read the Output**

Observe how the response is different from the weak prompt output.

The refined prompt should produce an answer that is:

- More structured
- Easier for beginners
- More defensive
- More useful for awareness training
- Better aligned with the lab objective



## Learner Observations

Learners should write down their observations after testing both prompts.

Suggested observation questions:

1. Was the weak prompt response clear enough for beginners?
2. Did the weak prompt response include practical safety actions?
3. Was the refined prompt response better organized?
4. Did the refined prompt stay defensive and awareness-focused?
5. Which response would be more useful in a cybersecurity awareness session?
6. How did adding role, task, context, constraints, and output format improve the result?